

# **ARTIFICIAL INTELLIGENCE**

## **FAKE NEWS DETECTION USING NATURAL LANGUAGE PROCESSING (NLP) :**

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### **PHASE 3 – DEVELOPMENT PART 1 :**

#### **INTRODUCTION :**

Fake news has become a significant concern in today's digital age, with the potential to spread misinformation and influence public opinion. **Natural Language Processing (NLP)** and **deep learning** techniques are powerful tools for fake news detection . In the previous documentation AI\_PHASE2 we saw how to implement the Long – Short Term Memory (LSTM) model for the Fake News Detection project .

In this step let us start building the model by loading and preprocessing the data . We will use python for the loading and preprocessing of the data .

Several modules like pandas , tensorflow , numpy , keras and genism will be used in this process .

#### **DEVELOPMENT STEPS :**

In this development process we are going to import required modules , load the dataset and preprocess the data for the training process .

##### **1. IMPORTING LIBRARIES :**

First we need to import the required modules need to perform the loading and preprocessing of the data .

Before importing the modules we need to install the modules .

```
pip install --upgrade tensorflow
pip install plotly
pip install --upgrade nbformat
pip install nltk
pip install spacy
pip install WordCloud
pip install gensim
pip install jupyterthemes
```

The tensorflow module is used for the machine learning process . The nltk (Natural Language Toolkit) is a library for natural language processing .Spacy is a open source software library for advanced natura language processing .Gensim is an open source library for unsupervised topic modelling and natural language processing .

After installing the requied packages now we need to import the modules for our project .

```
import nltk
nltk.download('punkt')

import tensorflow as tf
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from wordcloud import WordCloud, STOPWORDS
import nltk
import re
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize, sent_tokenize
import gensim
from gensim.utils import simple_preprocess
from gensim.parsing.preprocessing import STOPWORDS
# import keras
from tensorflow.keras.preprocessing.text import one_hot, Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten, Embedding, Input, LSTM, Conv1D, MaxPool1D, Bidirectional
```

```
from tensorflow.keras.models import Model
```

These modules will be used throughout the development process from the data loading and preprocessing to training the model and model evaluation .

## 2. **LOADING THE DATASETS :**

Now we need to load the required dataset in our program .

We use the fake news dataset form the Kaggle for this project . The link for the dataset is

<https://www.kaggle.com/clmentbisailon/fake-and-real-news-dataset> .

The dataset is downloaded and the two files “True.csv” and “Fake.csv” are moved in the folder named “data” .

Now the pandas library is used to load the dataset into our program

```
df_true = pd.read_csv("data/True.csv")  
df_fake = pd.read_csv("data/Fake.csv")
```

Now the dataset is loaded into the program .Now we can check the data is loaded correctly or not with the following code .

```
df_true.head()
```

It displays the first 5 items of the dataset . Similarly we can do for the fake data .

```
df_fake.head()
```

It displays the first 5 items of the fake news dataset .

Now we need to set the label for the fake and the real news datas .

For true values :

```
df_true['isfake'] = 1
```

For false values :

```
df_fake['isfake'] = 0
```

Sample output for `df_fake.head()`:

|   | title                                            | text                                              | subject | date              |
|---|--------------------------------------------------|---------------------------------------------------|---------|-------------------|
| 0 | Donald Trump Sends Out Embarrassing New Year'... | Donald Trump just couldn t wish all Americans ... | News    | December 31, 2017 |
| 1 | Drunk Bragging Trump Staffer Started Russian ... | House Intelligence Committee Chairman Devin Nu... | News    | December 31, 2017 |
| 2 | Sheriff David Clarke Becomes An Internet Joke... | On Friday, it was revealed that former Milwauk... | News    | December 30, 2017 |
| 3 | Trump Is So Obsessed He Even Has Obama's Name... | On Christmas day, Donald Trump announced that ... | News    | December 29, 2017 |
| 4 | Pope Francis Just Called Out Donald Trump Dur... | Pope Francis used his annual Christmas Day mes... | News    | December 25, 2017 |

### 3. DATA CLEANING :

The next step is to clean the dataset by removing unnecessary columns and datas that are not relevant for the training .

First we need to concatenate the real and fake datas together .

```
df = pd.concat([df_true, df_fake]).reset_index(drop = True)
```

This will concatenate the fake datas at the end of the real datas.

Now we are going to remove the date column which is not necessary for the model training .

```
df.drop(columns = ['date'], inplace = True)
```

This will remove the date column from the dataset .

Next step is to combine the “title” field and the “text” field and add them in the “original” field .

```
df['original'] = df['title'] + ' ' + df['text']
```

This will now create a new column “original” with the text and the title field combined .

Now after cleaning the datas we need to preprocess the data.

#### 4. DATA PREPROCESSING :

Now we need to preprocess the dataset like removing the stopwords .

```
nltk.download("stopwords")
```

Next we need to obtain the additional stopwords from nltk .

```
from nltk.corpus import stopwords  
stop_words = stopwords.words('english')  
stop_words.extend(['from', 'subject', 're', 'edu', 'use'])
```

The next step is to remove the stop words and remove words with 2 or less characters using genism.

```
def preprocess(text):  
    result = []  
    for token in gensim.utils.simple_preprocess(text):  
        if token not in gensim.parsing.preprocessing.STOPWORDS and  
len(token) > 3 and token not in stop_words:  
            result.append(token)  
  
    return result
```

Now we apply this function to the dataframe .

```
df['clean'] = df['original'].apply(preprocess)
```

The original data is

```
df['original'][0]
```

**Output:**

'As U.S. budget fight looms, Republicans flip their fiscal script....'

Now the cleaned up data after removing the stopwords is

```
print(df['clean'][0])
```

**Output:**

['budget', 'fight', 'looms', 'republicans', 'flip', 'fiscal', 'script',]....

Next we need to obtain the total words present in the dataset .

```
list_of_words = []  
for i in df.clean:  
    for j in i:  
        list_of_words.append(j)
```

Find the length of the list of words

```
len(list_of_words)
```

Output : 9276947

Then find the total number of unique words

```
total_words = len(list(set(list_of_words)))
```

total\_words

Output : 108704

```
df['clean_joined'] = df['clean'].apply(lambda x: " ".join(x))
```

The output of the df:

|       | title                                             | text                                              | subject      | isfake | original                                          | clean                                             | clean_joined                                      |
|-------|---------------------------------------------------|---------------------------------------------------|--------------|--------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0     | As U.S. budget fight looms, Republicans flip t... | WASHINGTON (Reuters) - The head of a conservat... | politicsNews | 1      | As U.S. budget fight looms, Republicans flip t... | [budget, fight, looms, republicans, flip, fisc... | budget fight looms republicans flip fiscal scr... |
| 1     | U.S. military to accept transgender recruits o... | WASHINGTON (Reuters) - Transgender people will... | politicsNews | 1      | U.S. military to accept transgender recruits o... | [military, accept, transgender, recruits, mond... | military accept transgender recruits monday pe... |
| 2     | Senior U.S. Republican senator: 'Let Mr. Muell... | WASHINGTON (Reuters) - The special counsel inv... | politicsNews | 1      | Senior U.S. Republican senator: 'Let Mr. Muell... | [senior, republican, senator, mueller, washing... | senior republican senator mueller washington r... |
| 3     | FBI Russia probe helped by Australian diplomat... | WASHINGTON (Reuters) - Trump campaign adviser ... | politicsNews | 1      | FBI Russia probe helped by Australian diplomat... | [russia, probe, helped, australian, diplomat, ... | russia probe helped australian diplomat washin... |
| 4     | Trump wants Postal Service to charge 'much mor... | SEATTLE/WASHINGTON (Reuters) - President Donal... | politicsNews | 1      | Trump wants Postal Service to charge 'much mor... | [trump, wants, postal, service, charge, amazon... | trump wants postal service charge amazon shipm... |
| ...   | ...                                               | ...                                               | ...          | ...    | ...                                               | ...                                               | ...                                               |
| 44893 | McPain: John McCain Furious That Iran Treated ... | 21st Century Wire says As 21WIRE reported earl... | Middle-east  | 0      | McPain: John McCain Furious That Iran Treated ... | [mcpain, john, mccain, furious, iran, treated,... | mcpain john mccain furious iran treated sailor... |
| 44894 | JUSTICE? Yahoo Settles E-mail Privacv Class-ac... | 21st Century Wire says It s a familiar theme ...  | Middle-east  | 0      | JUSTICE? Yahoo Settles E-mail Privacv Class-ac... | [justice, yahoo, settles, mail, privacv, class... | justice yahoo settles mail privacv class actio... |

## 5. TOKENIZATION :

Now let us perform the tokenization

```
from sklearn.model_selection import train_test_split
```

```
x_train, x_test, y_train, y_test = train_test_split(df.clean_joined, df.isfake, test_size = 0.2)
```

```
from nltk import word_tokenize
```

```
tokenizer = Tokenizer(num_words = total_words)
```

```
tokenizer.fit_on_texts(x_train)
train_sequences = tokenizer.texts_to_sequences(x_train)
test_sequences = tokenizer.texts_to_sequences(x_test)
;
```

Now we can print the train\_sequences:

```
print("The encoding for document\n",df.clean_joined[0],"\n is :  
",train_sequences[0])
```

Output :

The encoding for document  
budget fight looms republicans flip fiscal script washington.....  
[2817, 8135, 2981, 4196, 666, 1, 6236,541 ,27, 2444 .....]

```
padded_train = pad_sequences(train_sequences,maxlen = 40, padding =  
'post', truncating = 'post')  
padded_test = pad_sequences(test_sequences,maxlen = 40, truncating =  
'post')
```

```
for i,doc in enumerate(padded_train[:2]):  
print("The padded encoding for document",i+1," is : ",doc)
```

Output :

The padded encoding for document 1 is : [ 2817 8135 2981 4196 666  
1 6236 541 27 2444 1713 205  
2817 8135 5753 666 6236 10 1 45 541 1 2160 43225  
3477 183 1311 270 8135 641 1256 2444 17467 183 623 3126  
1240 19 1240 3836]  
The padded encoding for document 2 is : [ 133 2174 1732 2842 370  
120 442 226 9784 16805 17154 8  
133 8685 1732 322 11 370 1640 5264 4644 1808 442 226  
13539 942 1375 12180 390 90 1956 211 357 3549 2 11  
705 975 1956 743]

Thus the data is now converted into numbers using the tokenizer  
the next step is to train the model and evaluate it that will be done in the  
next phase project .